# SIMATS ENGINEERING
## ASSESSMENT TOOL 1

**NAME:** Siva Dakshitha K **REG.NO:**192425395

**COURSE NAME:** CSA1603 **COURSE CODE:** Data Warehousing and Data Mining

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**COURSE OUTCOME COVERED**

**CO1:** Design data warehouse architectures with appropriate dimensional models for effective decision support systems. BL1, BL2
Assessment Tool Weightage: 40%

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**QUESTIONS: 4** **Total Marks:20**

Annexure A
**MCQ TEST**

| *Criteria* | **Conceptual Understanding(3)** | **Accuracy of Answer(2.5)** | **Application of Knowledge(2)** | **Completeness of Response(1.5)** | **Clarity & Presentation(1)** | ***Total(10)*** |
|---|---|---|---|---|---|---|
| *Weightage* | *30%* | 25% | 20% | 15% | 10% | *100%* |
| *Q1* | | | | | | |
| *Q2* | | | | | | |
| *Q3* | | | | | | |
| *Q4* | | | | | | |

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***Code of Conduct:***

*I _Siva Dakshitha K / 192425395_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
*I understand that any violation of academic integrity rules will result in disciplinary action.*

Siva Dakshitha K
***Signature of the student:***

SIMATS ENGINEERING
Engineer to Excel

SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

# Beyond the Cube: The Secret Engine Behind Smart Business Decisions

CSA1603-Data Warehousing and Data Mining


**Faculty:**
Dr. D. Beulah David

**Presented By:**
Siva Dakshitha K


Engineer to Excel

# Introduction

**Why Do We Need Data Cube Computation?**

- Data is generated every second by businesses.
- Analyzing large amounts of data can be slow.
- **Data Cube Computation** organizes data for faster analysis.
- It stores data in **multiple dimensions** such as **Product, Time, and Location**.
- It helps users retrieve information quickly.
- It is widely used in **Data Warehousing** and **OLAP** for better decision-making.

**Example**

- Sales of Laptops in Chennai during January 2026

# What is a Data Cube?

**Definition**

A **Data Cube** is a multidimensional structure that organizes data for fast analysis.

**Three Dimensions**

- 📦 Product
- 🌍 Location
- 📅 Time

**Example:**

| Product | Location | Time | Sales |
| --- | --- | --- | --- |
| Laptop | Chennai | Jan 2026 | ₹18 lakhs |


**Benefits**

•Faster Queries

•Better Analysis

•Easy Decision Making

# Data Cube Computation Methods

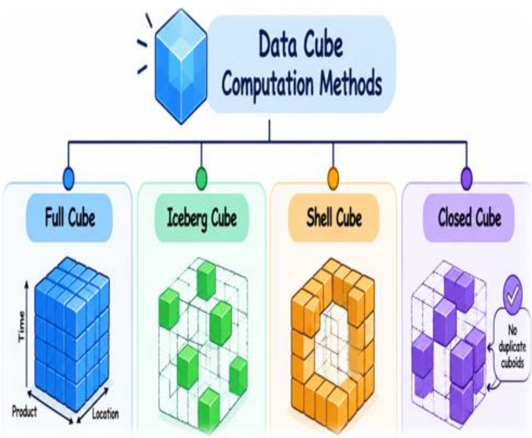
**Four Important Methods**

❖ **Full Cube**
- oComputes every possible combination.
- oFastest query performance.
- oRequires more storage.

❖ **Iceberg Cube**
- oStores only important values above a threshold.
- oSaves memory.
- oBest for large datasets.

❖ **Shell Cube**
- oComputes only selected combinations.
- oFlexible and efficient.

❖ **Closed Cube**
- oRemoves duplicate information.
- oReduces storage space.



# Real-Life Applications

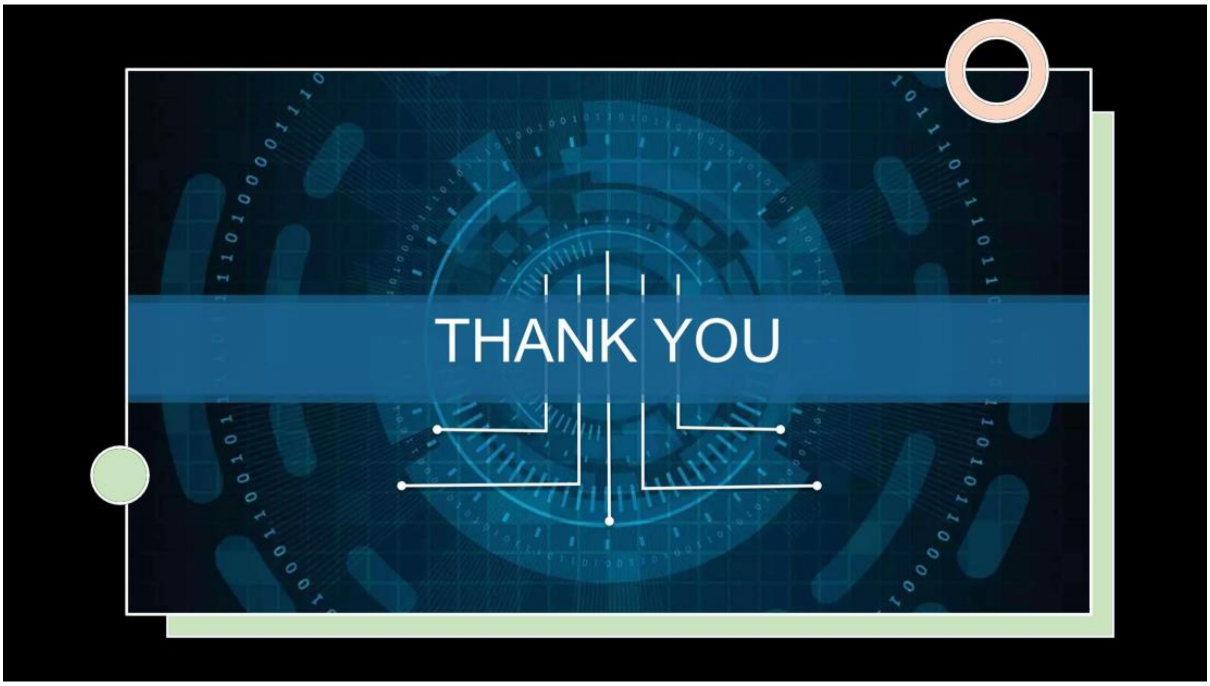
- ❖ **E-commerce**
  - Product recommendations
  - Customer purchase analysis
- ❖ **Banking**
  - Fraud detection
  - Financial reporting
- ❖ **Healthcare**
  - Patient trend analysis
  - Disease monitoring
- ❖ **Business Intelligence**
  - Sales forecasting
  - Performance dashboards

# Conclusion

- Data Cube Computation organizes data into multiple dimensions.
- It enables **fast and efficient query processing**.
- Different computation methods optimize **speed, storage, and performance**.
- It is a key component of **Data Warehousing** and **OLAP**.
- Businesses use Data Cube Computation to make **faster and smarter decisions**.


**Key Takeaway**

"Data Cube Computation transforms complex data into meaningful insights, helping organizations make better decisions with speed and accuracy."




**RUBRICS FOR CALCULATION**

| Criteria | Weightage | Level 4 – Exemplary (5 Marks) | Level 3 – Proficient (4 Marks) | Level 2 – Developing (2–3 Marks) | Level 1 – Beginning (0–1 Mark) |
|---|---|---|---|---|---|
| **Conceptual Understanding** | **30** | Demonstrates clear and complete understanding of the concept | Good understanding with minor gaps | Basic understanding with some misconceptions | Poor or incorrect understanding |
| **Accuracy of Answer** | **25** | Completely correct answer with no errors | Mostly correct with minor errors | Partially correct with noticeable errors | Incorrect or irrelevant answer |
| **Application of Knowledge** | **20** | Correctly applies concepts to solve the question/problem | Applies concepts with minor mistakes | Limited or partially correct application | No or incorrect application |
| **Completeness of Response** | **15** | Covers all required parts of the question | Covers most parts with minor omissions | Covers only some parts | Incomplete or missing response |
| **Clarity & Presentation** | **10** | Clear, concise, and well-organized answer | Understandable with minor issues | Somewhat unclear or poorly structured | Difficult to understand |